

# Nathan Kwan-Ho Ng

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## EDUCATION

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### Ph.D. in Computer Science

10/2019 – 09/2026 (Expected)

*University of Massachusetts Amherst*

Advisor: Prof. Prashant Shenoy

### B.Sc. in Computer Science, First Class Honors

09/2014 – 06/2018

*Hong Kong Baptist University*

Advisor: Prof. Byron Choi

## RESEARCH INTERESTS

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Systems for machine learning, distributed systems, edge and cloud computing.

## PUBLICATIONS & PREPRINTS

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**Tuning the Tail Latency of Distributed Queries Using Replication.** Nathan Ng, Hung Le, Prashant Shenoy, Marco Serafini. Under review at the *Distributed and Parallel Databases Journal (DAPD)*. arXiv Preprint [arXiv:2212.10387](https://arxiv.org/abs/2212.10387). Pages 1-12.

**Collaborative Processing for Multi-Tenant Inference on Memory-Constrained Edge TPUs.** Nathan Ng, Walid Hanafy, Prashanthi Kadambi, Balachandra Sunil, Ayush Gupta, David Irwin, Yogesh Simmhan, Prashant Shenoy. In *Proceedings of the International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT'26)*. Pages 1-8.

**To Offload or Not To Offload: Model-driven Comparison of Edge-native and On-device Processing In the Era of Accelerators.** Nathan Ng, David Irwin, Ananthram Swami, Don Towsley, Prashant Shenoy. In *Proceedings of the ACM/SPEC International Conference on Performance Engineering (ICPE'26)*. Pages 333-340.

**TailClipper: Reducing Tail Response Time of Distributed Services Through System-Wide Scheduling.** Nathan Ng, Abel Souza, David Irwin, Don Towsley, Prashant Shenoy. In *Proceedings of the ACM Symposium on Cloud Computing (SoCC'24)*. Pages 398-414.

**Collaborative Inference in Resource-Constrained Edge Networks: Challenges and Opportunities.** Nathan Ng, Abel Souza, Suhas Diggavi, Tarek Abdelzaher, Don Towsley, Prashant Shenoy. In *Proceedings of the IEEE Military Communications Conference (MILCOM'24 IoT for Adv. Env. Workshop)*. Pages 1-6.

**Unlocking Efficiency: Understanding End-to-End Performance in Distributed Analytics Pipelines.** Abel Souza, Nathan Ng, Tarek Abdelzaher, Don Towsley, Prashant Shenoy. In *Proceedings of the IEEE Military Communications Conference (MILCOM'23 IoT for Adv. Env. Workshop)*. Pages 150-155.

**FGreat: Focused Graph Query Autocompletion.** Nathan Ng, Peipei Yi, Zhiwei Zhang, Byron Choi, Sourav S Bhowmick, Jianliang Xu. In *Proceedings of the IEEE International Conference on Data Engineering (ICDE'19)*. Pages 1956-1959.

**Answering the Why-Not Questions of Graph Query Autocompletion.** Guozhong Li, Nathan Ng, Peipei Yi, Zhiwei Zhang, Byron Choi. In *Proceedings of the International Conference on Database Systems for Advanced Applications (DASFAA'18)*. Pages 332-341.

## PATENTS

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**Dynamic user profile management.** Nathan Ng, Tung Mai, Thomas Greger, Kelly Quinn Nicholes, Antonio Cuevas, Saayan Mitra, Somdeb Sarkhel, Anup Bandigadi Rao, Ryan A Rossi, Viswanathan Swaminathan, Shivakumar Vaithyanathan. *US Patent US12,339,916B2*. Published June 2025.

**Dynamic user profile projection.** Nathan Ng, Tung Mai, Thomas Greger, Kelly Quinn Nicholes, Antonio Cuevas, Saayan Mitra, Somdeb Sarkhel, Anup Bandigadi Rao, Ryan A Rossi, Viswanathan Swaminathan, Shivakumar Vaithyanathan. *US Patent US121,308,76B2*. Published Oct 2024.

## AWARDS AND HONORS

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|-----------------------------------------------------------------------------------------------------------------|-------------|
| <b>Donald F. Towsley Graduate Scholarship in Computer Science</b><br><i>University of Massachusetts Amherst</i> | 2026        |
| <b>James Kurose Scholarship in Computer Science</b><br><i>University of Massachusetts Amherst</i>               | 2020        |
| <b>Undergraduate Overseas Summer Research Scholarship</b><br><i>Hong Kong Baptist University</i>                | 2018        |
| <b>Outstanding Student Scholarship</b><br><i>Hong Kong Baptist University</i>                                   | 2017        |
| <b>President's Honor Roll</b><br><i>Hong Kong Baptist University</i>                                            | 2016 - 2018 |

## RESEARCH EXPERIENCE

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**Research Assistant, University of Massachusetts Amherst** 09/2022 - Present  
Advisor: Prof. Prashant Shenoy

- Developed SwapLess, a system enabling multi-tenant collaborative inference on memory-constrained Edge TPUs.
- Developed an analytic-model-driven resource manager for adaptive inference offloading in edge environments.
- Developed TailClipper, a queueing-theory-inspired scheduler that reduces tail latency of distributed applications.

**Incoming Research Scientist Intern, Adobe** 06/2026 - 09/2026 (Expected)  
Mentor: Dr. Saayan Mitra

- Developing privacy-preserving LLM inference offloading strategies within Adobe Acrobat.

**Research Scientist Intern, Adobe** 05/2022 - 08/2022  
Mentor: Dr. Tung Mai

- Developed an Observability Data Management System deployed in production for Adobe engineering teams.
- Developed a URL knowledge extraction system using Random Forest to improve metrics reporting.

**Data Science & Machine Learning Intern, Adobe** 05/2021 - 12/2021  
Mentor: Dr. Tung Mai

- Proposed cache admission algorithm using a predictive statistical model to forecast customer revisits.
- Proposed dynamic Cloud-Edge user profile replication algorithms to maximize cache hit rates.

**Student Research Collaborator, Meta** 01/2021 – 05/2021  
Mentor: Dr. Ritwik Yadav

- Benchmarked performance of RocksDB and LiveGraph using LinkBench on AWS EC2 instances.
- Collaborated on a potential deployment of LiveGraph as a Database-as-a-Service model at Meta.

**Research Assistant, University of Massachusetts Amherst** 09/2019 – 05/2022  
Advisor: Prof. Marco Serafini

- Proposed data replication algorithms to satisfy user-defined tail latency bounds for distributed queries.
- Developed a distributed graph query processing middleware that coordinated query execution across nodes.

**Research Assistant, Hong Kong Baptist University** 09/2018 – 08/2019

Advisor: Prof. Byron Choi

- Proposed algorithms for efficient indexed maximum common subgraph mining in graph databases.

**Undergraduate Student Researcher, Hong Kong Baptist University** 09/2017 – 08/2018

Advisor: Prof. Byron Choi

- Proposed ranking and search algorithms to address the ‘why-not’ questions in graph query autocompletion.

**Undergraduate Research Intern, Nanyang Technological University** 05/2017 – 08/2017

Advisor: Prof. Sourav S Bhowmick

- Developed FGreat, an interactive graph query autocompletion system that assists query formulation.

## TEACHING EXPERIENCE

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**CS 377: Operating Systems, Undergraduate teaching assistant** Spring 2025

*University of Massachusetts Amherst*

- Led lab sessions for 50 undergraduates covering OS concepts and hands-on assignments.

**CS 677: Distributed & Operating Systems, Graduate teaching assistant** Spring 2023, 2024

*University of Massachusetts Amherst*

- Held office hours, graded coursework, and provided Piazza support for students.

**CS 532: Systems for Data Science, Graduate teaching assistant** Fall 2019, 2022, 2023

*University of Massachusetts Amherst*

- Designed and led lab sessions on Hadoop and Spark for 120 graduate students.

**CS 2007: Object-Oriented Programming, Undergraduate teaching assistant** Spring 2018

*Hong Kong Baptist University*

- Assisted students with hands-on debugging and Java programming exercises.

**CS 1005: Essence of Computing, Undergraduate teaching assistant** Fall 2017

*Hong Kong Baptist University*

- Assisted freshman students with programming exercises and debugging.

## MENTORING

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Sribatscha Maharana, B.Sc. and MS in Computer Science (UMass’27) Spring 2026

Naga Surya Prathik Vunnava, MS in Computer Science (UMass’26) Spring 2026

Balachandra Devarangadi Sunil, MS in Computer Science (UMass’26) Fall 2025

Ayush Gupta, MS in Computer Science (UMass’26) Fall 2025

Khai Nguyen, MS in Computer Science (UMass’25) Spring 2025

Tanay Joshi, MS in Computer Science (UMass’25) Spring 2025

Rishi Patel, B.Sc. in Computer Science (UMass’25) Spring 2025

## SERVICE & OUTREACH

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**External Reviewer** Fall 2025

*ACM/IEEE Symposium on Edge Computing (SEC’25)*

**Shadow PC member** Fall 2025

*The European Conference on Computer Systems (EuroSys’26)*

## **REFERENCES**

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**Prashant Shenoy**

Distinguished Professor

*University of Massachusetts Amherst*

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**Don Towsley**

Distinguished Professor

*University of Massachusetts Amherst*

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**David Irwin**

Professor

*University of Massachusetts Amherst*

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**Hui Guan**

Assistant Professor

*University of Massachusetts Amherst*

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**Tung Mai**

Senior Research Scientist

*Adobe*

tumai@adobe.com